

AI TOOLS FOR RESEARCH

LAB EXERCISE SHEET

Scientific Writing Using Prism and LaTeX

Convert Your Own Research Paper into a Complete LaTeX Manuscript

Prism	https://prism.openai.com
Overleaf	https://www.overleaf.com

Student Name: _____ Roll No.: _____

Selected Paper Title: _____

1. Aim

To learn AI-assisted scientific writing and LaTeX document preparation by converting an existing research paper authored or co-authored by the student into a complete, structured, compilable LaTeX manuscript using Prism. The final project will be exported to Overleaf to test LaTeX portability and independent compilation.

2. Learning Outcomes

- Create and organize a scientific manuscript in a LaTeX project.
- Use Prism to assist with text-to-LaTeX conversion, equations, tables, figures, citations, and debugging.
- Use labels and cross-references correctly instead of hard-coded numbers.
- Verify that AI-assisted conversion preserves scientific meaning, numerical results, and citations.
- Export a LaTeX project and validate it independently in Overleaf.
- Maintain a prompt log and critically reflect on the benefits and limitations of AI-assisted scientific writing.

3. Input Required Before the Lab

- ☐ One research paper authored or co-authored by you, in PDF/Word/Google Docs or another usable format.
- ☐ Original figure files, wherever available.
- ☐ Tables and mathematical equations from the paper.
- ☐ Complete reference list or bibliography information.
- ☐ Access to Prism and Overleaf using the URLs given on the cover page.

4. Important Rule

This is a conversion and scientific document preparation exercise, not a paper-generation exercise. Do not use AI to invent new results, modify numerical values, create unsupported citations, or change the scientific conclusions of your original paper.

PART A. Start the Prism Project

Step 1. Open Prism and create a new project

1. Open Prism using <https://prism.openai.com>.
2. Sign in with the account available to you.
3. Create a new LaTeX project and give it a meaningful name, for example: Paper_Title_LaTeX.
4. Create or identify the main LaTeX source file, normally main.tex.
5. Compile once before adding the full manuscript so that you know the basic project works.

Suggested Prism prompt: *“Create a clean LaTeX article skeleton for a scientific research paper. Include title, authors, abstract, keywords, Introduction, Related Work, Methodology, Experimental Setup, Results and Discussion, Conclusion, and References. Do not generate scientific content.”*

Step 2. Create the minimum LaTeX structure

```
\documentclass{article}
\usepackage{graphicx}
\usepackage{amsmath}
\usepackage{booktabs}
\usepackage{cite}

\title{Your Research Paper Title}
\author{Author Names}

\begin{document}
\maketitle

\begin{abstract}
% Insert original abstract
\end{abstract}

\section{Introduction}
\section{Related Work}
\section{Methodology}
\section{Experimental Setup}
\section{Results and Discussion}
```

```
\section{Conclusion}

\bibliographystyle{plain}
\bibliography{references}
\end{document}
```

- ☐ The skeleton compiles successfully in Prism.
- ☐ The generated PDF shows the title and section headings.

PART B. Convert the Original Paper into LaTeX

Step 3. Transfer title, authors, affiliations, abstract, and keywords

6. Copy the title exactly from your paper.
7. Enter all author names and affiliations exactly.
8. Insert the original abstract.
9. Insert the original keywords.
10. Compile and compare the output with the original paper.

Suggested Prism prompt: *“Convert the following text into clean LaTeX. Preserve the exact wording, terminology, scientific meaning, numerical values, and citations. Do not add, remove, or reinterpret any scientific claim.”*

Item	Verified?	Observation
Title and author information preserved	Yes/ No	
Abstract meaning preserved	Yes/ No	
Numerical values preserved	Yes/ No	
No new claim introduced	Yes/ No	
Keywords preserved	Yes/ No	

PART C. Convert All Scientific Sections

Step 4. Convert the Introduction and remaining sections

11. Transfer the Introduction into Prism in manageable blocks.
12. Convert the text into LaTeX without changing the scientific content.
13. Use section and subsection commands only where they reflect the original manuscript structure.
14. Repeat the procedure for Related Work, Methodology, Dataset, Experimental Setup, Results, Discussion, Limitations, and Conclusion, wherever applicable.
15. Compile after every major section rather than waiting until the entire paper has been transferred.

Suggested Prism prompt: *“Convert this section into properly structured LaTeX. Preserve paragraph order, citations, terminology, equations, and all technical details. Do not summarize and do not introduce new content.”*

```
\section{Methodology}
\label{sec:methodology}

\subsection{Data Preparation}
...

\subsection{Proposed Method}
...
```

PART D. Equations and Mathematical Notation

Step 5. Convert equations to LaTeX

16. Identify every equation or important mathematical expression in the original paper.
17. Convert each equation into LaTeX mathematical notation.
18. Verify every operator, symbol, subscript, superscript, fraction, summation, Greek symbol, and variable.

19. Add a label to equations that are referenced in the text.
20. Use LaTeX cross-references instead of manually typing equation numbers.

Suggested Prism prompt: “Convert the following equation into valid LaTeX. Preserve the mathematics exactly. Then list the meaning of each variable and flag any ambiguity instead of guessing.”

```
\begin{equation}
RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}
\label{eq:rmse}
\end{equation}
```

The error is computed using Equation~\ref{eq:rmse}.

Equation Check	Status
Mathematics matches original	
Variable notation consistent	
Equation label added	
Cross-reference works	

PART E. Tables

Step 6. Recreate tables in LaTeX

21. Select every important table from the original paper. At minimum, complete one full table during the lab.
22. Provide the table data to Prism and ask for LaTeX table code.
23. Compare every cell with the original paper before accepting the output.
24. Add a caption and label, then reference the table from the text.

Suggested Prism prompt: “Convert this table into publication-quality LaTeX. Preserve every row, column heading, unit, decimal value, and result exactly. Do not round, reorder, or reinterpret any value.”

```
\begin{table}[ht]
\centering
\caption{Performance comparison of different models}
\label{tab:results}
\begin{tabular}{lccc}
\toprule
Model & Accuracy & Precision & F1-score \\
\midrule
Model A & 0.91 & 0.89 & 0.90 \\
Model B & 0.93 & 0.92 & 0.92 \\
\bottomrule
\end{tabular}
\end{table}
```

The comparative results are shown in Table~\ref{tab:results}.

PART F. Figures

Step 7. Add figures and cross-references

25. Upload original figure files to the Prism project. Use meaningful file names.
26. Insert each figure with an appropriate width, caption, and label.
27. Reference the figure using its LaTeX label.
28. Compile and check figure quality, placement, caption, and numbering.

```
\begin{figure}[ht]
\centering
\includegraphics[width=0.8\linewidth]{proposed_framework.png}
\caption{Overview of the proposed framework.}
\label{fig:framework}
\end{figure}
```

As shown in Figure~\ref{fig:framework}, ...

Suggested Prism prompt: “Check this figure environment for correct LaTeX syntax, captioning, file path, sizing, and crossreferencing. Do not change the scientific caption unless there is a clear grammatical error.”

PART G. Citations and Bibliography

Step 8. Create references.bib and convert citations

29. Create a bibliography file named references.bib.
30. Enter the bibliographic details of the references used in the original paper.
31. Replace manually typed citation numbers with LaTeX citation commands.
32. Compile the bibliography and verify that every in-text citation resolves correctly.
33. Compare the generated reference list against the original paper.

Suggested Prism prompt: “Create a BibTeX entry from the bibliographic information below. Use only the information supplied. Do not invent DOI, volume, issue, page numbers, publisher, or venue details.”

```
@article{smith2025,
  author = {John Smith and Alice Brown},
  title = {Example Research Paper},
  journal = {Journal of Artificial Intelligence},
  year = {2025},
  volume = {10},
  number = {2},
  pages = {100--115}
}
```

Verification rule: AI-generated or AI-reformatted references must be checked against the original publication information before submission.

PART H. Cross-Referencing

Step 9. Replace hard-coded numbers with labels and references

Object	Label Example	Reference Example
Section	\label{sec:methodology}	Section~\ref{sec:methodology}
Figure	\label{fig:framework}	Figure~\ref{fig:framework}
Table	\label{tab:results}	Table~\ref{tab:results}
Equation	\label{eq:rmse}	Equation~\ref{eq:rmse}

- ☐ At least one section reference works.
- ☐ At least one figure reference works.
- ☐ At least one table reference works.
- ☐ At least one equation reference works.

PART I. Prism AI for LaTeX Debugging

Step 10. Diagnose and document at least three LaTeX problems

Possible problems include undefined citations, undefined references, missing braces, missing packages, figure-path errors, table overflow, equation syntax errors, and bibliography compilation problems.

Suggested Prism prompt: “Explain this LaTeX compilation error. Identify the likely cause, show the minimum correction required, and explain why the correction works. Do not rewrite unrelated parts of the document.”

Error / Problem	Likely Cause	Prism Suggestion	Final Correction
1.			
2.			

3.			
----	--	--	--

PART J. AI-Assisted Scientific Writing

Step 11. Improve one paragraph without changing the science

34. Select one paragraph from your original paper.
35. Save the original paragraph as Version A.
36. Use Prism to improve clarity and scientific style. Save the output as Version B.
37. Review every change and create your own final Version C.

Suggested Prism prompt: *"Improve the clarity, conciseness, grammar, and scientific tone of this paragraph. Preserve all technical terminology, citations, numerical values, causal claims, limitations, and scientific meaning. Do not introduce new claims."*

Version	What to Submit
A	Original paragraph
B	Prism-assisted paragraph
C	Your final verified paragraph

Brief analysis: Identify what Prism improved, what you rejected or modified, and whether any edit could have changed the scientific meaning.

PART K. Consistency Check

Step 12. Ask Prism to report inconsistencies

Suggested Prism prompt: *"Review this manuscript for consistency in terminology, abbreviations, variable names, mathematical notation, dataset names, model names, capitalization, units, figure references, table references, and citation style. Report inconsistencies only. Do not rewrite the manuscript."*

Inconsistency Found	Original Usage	Corrected Usage

PART L. Export to Overleaf and Validate Portability

Step 13. Export the Prism project

38. Export or download the complete LaTeX project from Prism.
39. Confirm that the exported package includes .tex files, references.bib, figures, and any other required files.
40. Keep the exported files unchanged for the first Overleaf compilation attempt so that you can genuinely test portability.

Step 14. Import into Overleaf

41. Open <https://www.overleaf.com>.
42. Create a new project and upload/import the complete Prism project.
43. Select the correct main .tex file if Overleaf does not detect it automatically.
44. Compile the project.
45. If compilation fails, identify the cause before making corrections.

Component	Prism	Overleaf	Difference / Action Required
Compilation			
Document structure			
Equations			

Tables			
Figures			
Citations			
References			
Cross-references			

Suggested Prism prompt: *“My LaTeX project compiles in Prism but not in Overleaf. Based on the following error log and project structure, identify the most likely portability issue and suggest the smallest fix. Do not change scientific content.”*

PART M. Prompt Log

Step 15. Maintain evidence of AI usage

Record at least five meaningful prompts. The prompts should show that you gave Prism precise, task-specific instructions rather than asking it simply to “fix” or “write” the document.

No.	Task	Prompt Used	Accepted / Modified / Rejected
1	Text-to-LaTeX		
2	Equation conversion		
3	Table conversion		
4	Debugging		
5	Scientific writing		
6	Consistency check		

PART N. Final Verification Against the Original Paper

Step 16. Complete the verification checklist

- ☐ Title, authors, affiliations, abstract, and keywords match the original paper.
- ☐ All major sections of the original paper are present.
- ☐ All numerical results and dataset statistics are unchanged.
- ☐ All equations are mathematically equivalent to the original.
- ☐ All tables contain the correct values, units, labels, and captions.
- ☐ All figures are present and correctly referenced.
- ☐ All citations resolve and no reference has been fabricated.
- ☐ All section, equation, table, and figure cross-references work.
- ☐ The Prism project compiles without errors.
- ☐ The exported project has been tested in Overleaf.

PART O. Reflection

Step 17. Write a 200-word reflection

46. Which part of converting the paper into LaTeX was easiest?
47. Which part required the most manual effort?
48. Which Prism feature was most useful and why?
49. Did Prism introduce any technical, citation, numerical, or scientific errors?
50. How much manual correction was required?
51. Did the same project compile in Overleaf without modification?
52. What precautions should researchers take when using AI for scientific writing?

5. Final Submission

Submit one folder using the following naming convention: **StudentName_RollNumber**

```
StudentName_RollNumber/
|
|-- Original_Paper.pdf
|-- Prism_Final_Paper.pdf
|-- Overleaf_Final_Paper.pdf
|-- main.tex
|-- references.bib
|-- Figures/
|-- Prompt_Log.pdf or .docx
|-- Error_Log.pdf or .docx
`-- Reflection.pdf or .docx
```

6. Minimum Requirements

- ☐ Complete conversion of your own research paper into LaTeX.
- ☐ At least one equation converted and cross-referenced, if the paper contains equations. If the paper contains no equations, state this explicitly.
- ☐ At least one table converted to LaTeX if the paper contains tables.
- ☐ At least one figure inserted and cross-referenced, if the paper contains figures.
- ☐ At least five verified references managed through LaTeX/BibTeX.
- ☐ At least four working cross-references across sections, tables, figures, and/or equations.
- ☐ At least three documented LaTeX errors or formatting problems and their fixes.
- ☐ At least five meaningful Prism prompts in the prompt log.
- ☐ One original versus Prism-assisted paragraph comparison.
- ☐ Prism-to-Overleaf portability test.

7. Expected Workflow

Own Research Paper → **Prism** → **Structured LaTeX** → **Equations / Tables / Figures** → **Citations** → **Debugging** → **Verification** → **Export** → **Overleaf** → **Final Compilable Manuscript**

Key Principle: Use AI to reduce the mechanical effort of scientific document preparation, but retain human responsibility for scientific correctness, numerical accuracy, citation validity, interpretation, and final authorship.

Useful Links: [Prism](#) | [Overleaf](#)